

```
[2]: import pandas as pd

# Student Data
data = {
    "Name": ["Ali", "Rohan", "John", "Emma", "David", "Eisha", "Rahul", "Priya", "Sophia", "Neha"],
    "Math": [85, 92, 70, 88, 60, 95, 78, 82, 90, 76],
    "Science": [78, 88, 65, 90, 72, 91, 80, 85, 89, 74],
    "English": [90, 95, 75, 85, 68, 93, 79, 87, 92, 80]
}

# Create DataFrame
df = pd.DataFrame(data)

# Save dataset as CSV
df.to_csv("students.csv", index=False)

print("===== STUDENT DATASET =====")
print(df)

# Total Marks
```

```
print("==== STUDENT DATASET ====")
print(df)

# Total Marks
df["Total"] = df["Math"] + df["Science"] + df["English"]

# Average Marks
df["Average"] = df["Total"] / 3

print("\n==== TOTAL & AVERAGE ====")
print(df[["Name", "Total", "Average"]])

# Top Performer
top_student = df.loc[df["Total"].idxmax()]

print("\n==== TOP PERFORMER ====")
print("Name:", top_student["Name"])
print("Total Marks:", top_student["Total"])

# Subject-wise Statistics
print("\n==== SUBJECT STATISTICS ====")

print("Math Average:", df["Math"].mean())
```



```
# Subject-wise Statistics
print("\n===== SUBJECT STATISTICS =====")

print("Math Average:", df["Math"].mean())
print("Science Average:", df["Science"].mean())
print("English Average:", df["English"].mean())

print("\nHighest Math Marks:", df["Math"].max())
print("Highest Science Marks:", df["Science"].max())
print("Highest English Marks:", df["English"].max())

print("\nLowest Math Marks:", df["Math"].min())
print("Lowest Science Marks:", df["Science"].min())
print("Lowest English Marks:", df["English"].min())

# Save final report
df.to_csv("student_report.csv", index=False)

print("\nAnalysis Completed Successfully!")
print("Files Created:")
print("1. students.csv")
print("2. student_report.csv")
```



Output:

```
===== STUDENT DATASET =====
      Name  Math  Science  English
0     Ali    85      78      90
1    Rohan    92      88      95
2     John    70      65      75
3     Emma    88      90      85
4    David    60      72      68
5     Eisha    95      91      93
6    Rahul    78      80      79
7    Priya    82      85      87
8   Sophia    90      89      92
9     Neha    76      74      80
```

```
===== TOTAL & AVERAGE =====
      Name  Total  Average
0     Ali    253  84.333333
1    Rohan    275  91.666667
2     John    210  70.000000
3     Emma    263  87.666667
4    David    200  66.666667
5     Eisha    279  93.000000
6    Rahul    237  79.000000
7    Priya    254  84.666667
8   Sophia    271  90.333333
```



```
8 Sophia 271 90.333333
9 Neha 230 76.666667
```

```
===== TOP PERFORMER =====
```

```
Name: Eisha
```

```
Total Marks: 279
```

```
===== SUBJECT STATISTICS =====
```

```
Math Average: 81.6
```

```
Science Average: 81.2
```

```
English Average: 84.4
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```
Highest Math Marks: 95
```

```
Highest Science Marks: 91
```

```
Highest English Marks: 95
```

```
Lowest Math Marks: 60
```

```
Lowest Science Marks: 65
```

```
Lowest English Marks: 68
```

```
Analysis Completed Successfully!
```

```
Files Created:
```

```
1. students.csv
```

```
2. student_report.csv
```

Brief Description:

Task 3: Data Analysis Project (Pandas) Performed data analysis using the Pandas library by importing, cleaning, processing, and analyzing datasets. Generated meaningful insights through data manipulation and summary statistics.